MYTHBUSTER: AUTOMATED FACT-CHECKING OF ARABIC DIABETES VIDEOS

MenaML 2026 at KAUST, Saudi Arabia

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ABSTRACT

Diabetes misinformation spreads rapidly in short Arabic videos and can lead to harmful health decisions. We present MythBuster, an automated pipeline that fact-checks diabetes-related claims in short Arabic videos. The system performs speech-to-text transcription, extracts checkable claims, retrieves supporting evidence from trusted medical websites and a research corpus (RAG), then generates evidence-grounded verdicts with concise medical reasoning in Arabic. Our experiments show that Whisper-Large provides the strongest Arabic transcription quality and that retrieval grounding substantially improves claim verification reliability.

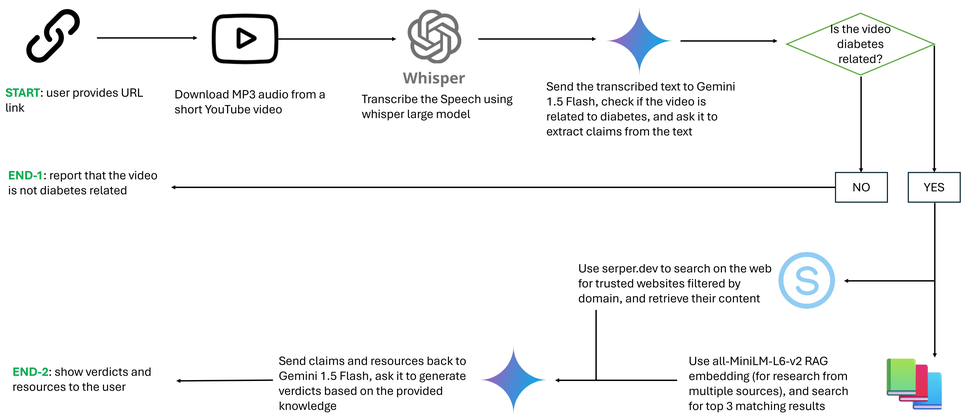
MOTIVATION

- Arabic short-form videos frequently include unverified diabetes advice and “miracle cure” narratives.
- Manual verification is slow; viewers need a fast, automated way to check claims.
- Most existing fact-checking systems and resources are optimized for English, not Arabic medical content.

CONTRIBUTIONS

- Contribution One: End-to-end system for automated fact-checking of short Arabic diabetes videos (URL → verdict).
- Contribution Two: Hybrid evidence retrieval: domain-filtered trusted web search + RAG over scientific literature (ChromaDB).
- Contribution Three: Structured outputs: extracted claims and verdicts (`TRUE` / `FALSE` / `PARTIALLY_TRUE` / `INSUFFICIENT_INFO`) with Arabic medical reasoning.

OVERVIEW

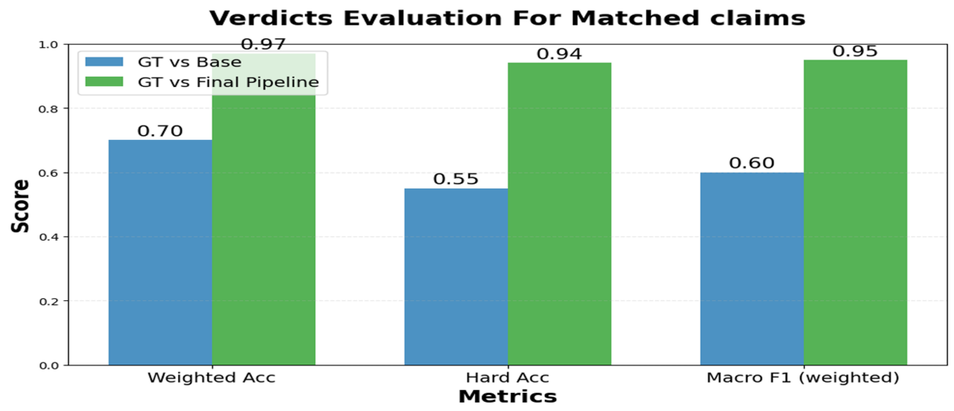


End-to-end pipeline: video ingestion, transcription, claim extraction, evidence retrieval (trusted web + RAG), and verdict generation.

RESULTS

Evaluation: We adopt an LLM-as-a-Judge protocol (GPT-5) to compare claim extraction and verdict quality across pipeline configurations.

- Whisper-Large outperforms Whisper-Medium, GPT-4o, and GPT-4o-mini in Arabic transcription.
- **Pipeline design > model size (core result):** With trusted web search and RAG, Gemini 1.5 Flash achieves near–GPT-5 verdict quality, while the same model without retrieval performs substantially worse.



FUTURE DIRECTIONS

- Direction 1: Expand beyond diabetes to other high-risk Arabic health misinformation topics.
- Direction 2: Improve evidence quality by expanding and systematically filtering trusted medical sources in the retrieval pipeline.
- Direction 3: Add confidence estimation to explicitly communicate uncertainty in generated verdicts.

DEMO & CODE

Demo

GitHub

REFERENCES

- OpenAI. *Whisper*.
- Chroma. *ChromaDB*.
- Google. *Gemini API*.
- Serper API.
- arXiv.
- yt-dlp.